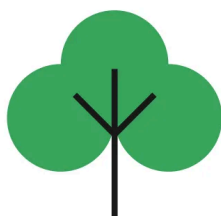




# **Yagé Botanicals Cloud Migration Proposal**



Yagé Botanicals

## **Network Modernization Proposal**

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## Step 1: Identify assets



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### Overview

Before Yagé Botanicals can move forward with any modernization efforts, we need a solid understanding of their current infrastructure. Right now, there's no comprehensive inventory of their hardware, software, or data. This is a big problem—not just for improving security, but also for making the company more appealing to buyers. A lack of visibility means it's harder to identify risks, vulnerabilities, or opportunities for improvement.

### Objectives:

1. Establish a clear picture of Yagé Botanicals' critical assets.
2. Highlight risks tied to outdated systems and hardware.
3. Create a foundation for improved cybersecurity and operational efficiency.

### Key Activities:

1. **Conduct a Full Hardware and Software Inventory:**
  - Use automated tools, like network scanners and asset discovery software, to map all hardware and software across the company's five sites. This will replace their current "manila folder" system and ensure all devices, from servers to employee devices, are accounted for.
  - Categorize and tag assets by type (e.g., workstations, servers, mobile devices) and usage (e.g., accounting, inventory management).
2. **Evaluate System Criticality and Dependencies:**

- Identify which systems are mission-critical, such as the servers hosting accounting and inventory applications. Note dependencies between systems—for example, how the accounting server relies on the file-sharing system.
  - Work with the IT team and key departments to document daily workflows, ensuring no essential systems are overlooked.
3. **Assess Cybersecurity Risks and Vulnerabilities:**
- Perform a basic vulnerability scan on all identified assets. This will help identify weak points, such as missing patches or unsupported software versions, that could be exploited by attackers.
  - Review the current state of Active Directory and network segmentation to uncover any glaring issues.

## **Approach**

Since the team has limited resources, I'll prioritize automation where possible, using tools like Nmap, Angry IP Scanner, or Spiceworks to streamline the discovery process. Additionally, I'll rely on input from staff who interact with these systems daily. This collaborative approach ensures we not only identify technical assets but also understand their business importance.

## **Identified Assets**

### **Hardware:**

- Fully used 12 servers, with 7 actively operational and others in varying states.
- UTM firewall at the home office (outdated, with subscription expired for 18 months).
- RODCs (Read-Only Domain Controllers) at each of the five sites to assist with logins.
- A mix of outdated workstations and personal devices employees bring to work.

### **Software:**

- Core systems for accounting and inventory management running on older versions of Windows.
- The company website, which is basic and hosted by a third party.

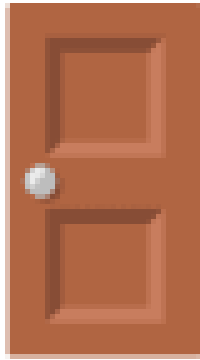
### **Data:**

- Customer purchase history, which is a key asset for the business.
- Inventory data and operational records, including vendor information.

***In conclusion, taking a full inventory of Yagé Botanicals' assets is the first big step toward modernizing their IT setup. By using smart tools and input from the team, we'll make sure***

*we're covering everything important. This step sets the stage for improving security, streamlining operations, and making the company more appealing to potential buyers.*

## Step 2: Implement basic security measures



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Based on my initial observations and the asset inventory we developed in Step 1, Yagé Botanicals needs foundational improvements to address gaps in their cybersecurity hygiene. These actions will help create a safer environment for daily operations while preparing for the larger infrastructure overhaul.

### **Proposed Actions:**

#### **Apply Critical Updates and Patches**

- I recommend starting with patching the outdated servers and workstations to reduce vulnerabilities.
- Priority should go to critical systems, like the accounting and inventory servers, to secure the company's core operations.
- Patching addresses known risks without requiring a major investment upfront.

#### **Enable and Enforce Multi-Factor Authentication (MFA)**

- As an initial step, I suggest implementing MFA for privileged accounts, like administrative users.
- This is a practical way to strengthen system access and prevent unauthorized logins.
- MFA can be gradually rolled out to all employees in later phases.

#### **Segment the Network with VLANs**

- To immediately reduce the risk of lateral movement in case of a breach, I recommend creating VLANs to isolate sensitive systems (e.g., servers, inventory data) from general user devices.
- This logical separation is cost-effective and aligns with best practices for internal threat protection.

### **Implement Endpoint Protection via a SaaS-Based MDM Solution**

- Personal devices connecting to the corporate network need to comply with a formal BYOD (Bring Your Own Device) security policy.
- Using a SaaS-based MDM (Mobile Device Management) tool can enforce security requirements like antivirus, firewalls, and secure configurations. This balances cost and security while respecting employee device privacy.

### **Renew the Firewall Subscription and Reconfigure Basic Rules**

- Reactivating the UTM firewall subscription is critical for regaining control over network perimeter defenses.
- Once reactivated, I suggest applying default-deny rules and limiting unnecessary inbound/outbound traffic to reduce exposure to external threats.

### **Launch a Basic Security Awareness Program**

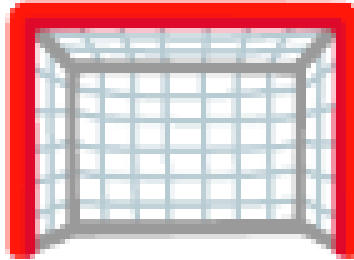
- Employees are often the first line of defense. A quick security training session can teach them how to identify phishing emails, use strong passwords, and safely handle company data.
- This can start as a one-time session and evolve into regular refreshers.

These steps represent an affordable, achievable starting point to improve Yagé Botanicals' cybersecurity hygiene. By focusing on practical, high-impact actions, we can create a solid foundation to build upon as the company transitions to the cloud.

### **Why These Actions Matter**

*Taking these deliberate actions now ensures that the systems and data we identified in Step 1 are immediately more secure. It also demonstrates to potential buyers that Yagé Botanicals is committed to modernizing their IT practices without overextending their resources.*

### Step 3: Strategize the migration



## Step 3: Strategize the migration.

Based on my research and understanding of Yagé Botanicals' challenges, I've outlined a step-by-step approach to modernize their infrastructure while balancing security, efficiency, and cost-effectiveness. This strategy aims to address the concerns of potential buyers and ensure the company is ready for future growth.

### 1. Tackle the Biggest Issues First

To start, focus on quick wins that will immediately improve operational efficiency and security:

- Upgrade the Active Directory domain controller to Windows Server 2022 on supported hardware. This step replaces outdated systems and enhances network authentication processes.
- Implement VLANs to segment the flat network, isolating critical systems like accounting and inventory from general user devices. This will limit lateral movement and improve internal security.
- Reinstate the UTM firewall subscription and configure it to enforce modern security policies, such as intrusion prevention and content filtering. This addresses the current lack of active firewall protection.

### 2. Migrate to a Hybrid Cloud Model

To modernize while keeping costs manageable, a phased migration approach is recommended:

- Begin by migrating less sensitive workloads — such as inventory management, email, and basic file sharing — to a public cloud provider like Microsoft Azure or AWS. This reduces the strain on on-premise resources and enhances scalability.

- Retain critical and sensitive systems, such as customer purchase history and operational records, on-premises for now, ensuring they are secured through updated configurations and monitoring.
- Leverage Microsoft Azure Active Directory (Azure AD) to integrate on-premises systems with cloud services, simplifying identity management across the hybrid environment.

### **3. Optimize Network and Connectivity**

Enhancing connectivity and reliability will significantly boost productivity:

- Replace the existing slow 5 Mbps sDSL connections with business-grade fiber internet for faster, more reliable bandwidth across all five locations.
- Utilize Azure VPN Gateway to securely connect on-premises systems with the cloud environment during and after migration, ensuring seamless operations.

### **4. Streamline IT Operations**

To support ongoing operations and reduce costs:

- Streamline the 12 servers by combining them into fewer, optimized machines. Phase out or repurpose any outdated or unnecessary hardware to cut down on maintenance costs.
- Implement an automated patch management solution to keep systems and software up-to-date without burdening the IT staff.
- Use a SaaS-based MDM solution to enforce security policies on employee-owned devices, enhancing endpoint protection while keeping costs low.

### **5. Engage Employees in the Transition**

To ensure a smooth transition and maintain operational uptime:

- Provide training for IT staff and key employees on managing and using hybrid cloud systems effectively.
- Conduct regular check-ins with department heads to identify any workflow disruptions and address them promptly.

***By following this roadmap, Yagé Botanicals will modernize their infrastructure in a way that balances buyer appeal, operational efficiency, and cost-effectiveness. This phased approach ensures the company is positioned for a successful sale while minimizing disruptions to its daily operations.***



## Step 4: Establish ongoing security monitoring



## Step 4: Establish ongoing security monitoring.

Yagé Botanicals has expressed concerns about keeping their systems running smoothly and protecting against security threats. It's a valid worry—cyber incidents can hurt a business's value and even lead to its downfall. With the company's plans to sell, now is the time to step up its cybersecurity game.

### **Recommended Course of Action: Hybrid SNOC**

Given Yagé's small IT team and tight budget, setting up a Hybrid Security and Network Operations Center (SNOC) is the most practical way forward. A hybrid SNOC combines network and security monitoring, helping Yagé address operational and security challenges at the same time. This approach uses the same tools for both, saving time and money.

### **Why a SNOC?**

- Buyers are looking for an efficient way to handle network and security operations.
- Combining these functions gives better visibility, cuts downtime, and lowers risks.

### **Immediate Steps for Implementation**

#### **1. Use SaaS-Based Tools for Monitoring**

Start with SaaS tools like Microsoft Sentinel or Splunk. These tools are easy to set up and will centralize network and security monitoring. They're also scalable, which is a big selling point for buyers.

## 2. Automate Routine Responses with SOAR

To lighten the IT team's load, implement a Security Orchestration, Automation, and Response (SOAR) solution. For example:

- Automatically respond to alerts like unusual logins or potential malware detections.
- Use pre-built playbooks to handle common issues, like quarantining suspicious devices.

## 3. Focus on Protecting Critical Data

The customer database and operational records are the backbone of the business. Set up real-time alerts for unauthorized access or suspicious activity targeting these systems.

## 4. Train the IT Team

Help the IT staff get comfortable with the new tools. A phased training plan will ensure smooth adoption without disrupting daily operations.

### Long-Term Monitoring Goals

- **Regular Threat Reviews:** Schedule periodic reviews to find and fix vulnerabilities before they become problems.
- **Track Performance:** Measure things like network uptime and how quickly incidents are resolved. These numbers will show buyers that Yagé is reliable.
- **Follow Best Practices:** Even though Yagé isn't under strict regulations, following recognized guidelines like CIS Controls will make the business more appealing.

### How This Plan Addresses Buyer Concerns

- **Stronger Security Posture:** Buyers will see that Yagé is serious about protecting its systems and data.
- **Improved Stability:** Better monitoring tools and processes will reduce downtime and keep things running smoothly.
- **Cost-Efficient Solutions:** SaaS tools and automation save money while modernizing operations.

*This approach keeps costs manageable and shows buyers that Yagé is ready to step into the modern age of IT security and reliability.*



## Step 5: Stabilize ongoing business operations



## Step 5: Stabilize ongoing business operations.

One of the biggest concerns for prospective buyers has been the outdated and disorganized IT environment at Yagé Botanicals. It's clear that stabilizing day-to-day operations is essential—not just for making the business more appealing but also for improving the company's efficiency and reliability in the short term.

Here are three key activities that will help Yagé modernize and bring structure to their business operations:

### 1. Develop and Enforce IT Policies and Procedures

Right now, Yagé's IT environment lacks clear rules and guidelines. Buyers are concerned about the risks that come with an informal approach to technology. To address this:

- Create a formal IT policy covering areas like acceptable use, password management, and device security.
- Document standard operating procedures (SOPs) for common tasks like onboarding employees, applying updates, and responding to incidents.
- Conduct regular training sessions to make sure staff understand and follow these policies.

**Why This Matters:** Buyers will see that the company is taking a professional, proactive approach to managing IT risks. It also prevents small issues from becoming larger problems.

### 2. Implement Basic IT Management Tools

Managing IT systems manually is inefficient and prone to errors. Introducing a few essential tools will streamline operations and improve reliability:

- **Remote Monitoring and Management (RMM):** Use tools like NinjaOne or Atera to track system performance and address issues before they escalate.
- **Patch Management:** Automate updates for operating systems and software to minimize vulnerabilities.
- **Centralized Documentation:** Move away from scattered notes and create a unified knowledge base that the IT team can reference.

**Why This Matters:** These tools will simplify IT management, reduce downtime, and demonstrate to buyers that Yagé has an organized, scalable approach to technology.

### **3. Strengthen Backup and Disaster Recovery Plans**

Yagé's lack of a proper backup system is a red flag for buyers. To stabilize operations and protect critical data:

- Use cloud-based backup solutions like Veeam or Acronis to ensure offsite, secure backups.
- Establish a disaster recovery plan that includes clear steps for restoring operations in case of an outage or cyber incident.
- Regularly test backups to make sure they work when needed.

**Why This Matters:** Reliable backups and a tested recovery plan reduce the risk of data loss and reassure buyers that the company can handle unexpected disruptions.

### **How These Steps Stabilize Operations**

By implementing clear policies, automating IT management, and protecting critical data, Yagé Botanicals can create a more stable and efficient business operation. Buyers will see a company that has addressed its weaknesses, making it a much more attractive investment.

***This approach balances immediate fixes with long-term improvements, ensuring Yagé Botanicals is ready for a smoother sale and a stronger future.***